

Schedule and Cost Performance Analysis

Capital Improvement Program — Q2 Program Review

DRAFT — Unedited Working Document — Prepared by Program Controls

1. Purpose

This report ~~has been prepared for purposes of summarizing~~ the current schedule and cost performance of the capital improvement program (the “Program”) as of the reporting date indicated on the cover of this document. ~~the~~ The intent being is to provide program leadership, as well as the various stakeholder departments participating in the Program, with a consolidated, portfolio-level view of how the individual projects ~~that comprise~~ comprising the Program are performing relative to their approved baseline schedules and budgets, ~~utilizing~~ the Earned Value Management (EVM) methodology ~~that has been adopted~~ as the standard performance measurement approach for ~~purposes of~~ Program reporting going forward. ~~it being noted~~ Note that not all of the projects ~~that comprise~~ comprising the Program had reached a sufficient level of design completion as of the reporting date to support meaningful EVM analysis, and ~~that those projects~~ have accordingly been excluded from the quantitative analysis presented here. ~~in, although they~~ They continue to be tracked qualitatively and will be incorporated into the quantitative reporting once sufficient baseline data becomes available.

2. Methodology

Earned Value Management, ~~for purposes of readers who may not be familiar with the methodology,~~ is a project management technique that integrates scope, schedule, and cost data ~~in order to~~ provide an objective measurement of project performance. ~~the~~ Its core metrics ~~of which~~ include Planned Value (PV), which represents the budgeted cost of work ~~that was~~ scheduled to be performed as of a given point in time; Earned Value (EV), which represents the budgeted cost of work ~~that has actually been~~ performed as of that same point in time; and Actual Cost (AC), which represents the actual cost ~~that has been~~ incurred ~~in~~ performing that work.

~~It being from these three base metrics that the~~ two primary performance indices are utilized in this report: ~~are derived, namely~~ the Schedule Performance Index (SPI), ~~calculated as EV divided by PV,~~ and the Cost Performance Index (CPI). ~~Both are derived from the three basic metrics defined above: PV, EV, and AC. SPI is calculated as EV divided by PV, and CPI is calculated as EV divided by AC, with an~~ An index value of 1.0 ~~indicating indicates~~ performance exactly in line with the baseline plan; values greater than 1.0 ~~indicating indicate~~ favorable performance; (ahead of schedule ~~in the case of for~~ SPI, ~~or~~ under budget ~~in the case of for~~ CPI), and values less than 1.0 indicating unfavorable performance. ~~It being is~~ generally accepted ~~within the program controls profession~~ that index values falling outside the range of approximately 0.90 to 1.10 warrant further management attention and, ~~in many cases,~~ often a formal variance explanation from the responsible project manager.

3. Portfolio-Level Findings

The Program is running meaningfully behind schedule and over budget, as reflected in a composite SPI of 0.87 and a composite CPI of 0.93 for the current reporting period. Both figures fall outside the general threshold range noted above, triggering management attention under the Program's

Commented [RS1]: I checked the numbers, and I calculated SPI of 0.89 and 0.94. They are close enough that it could be a result of rounding, but it's worth double checking the numbers.

reporting protocol. This is broadly consistent with anecdotal feedback from several project managers citing permitting delays and, in some cases, procurement lead times for long-lead equipment that now exceed the assumptions built into the original baseline schedules. This suggests that current market conditions for equipment procurement no longer match the conditions assumed when those schedules were developed. As of the current reporting period, the Program as a whole reflects a composite SPI of 0.87 and a composite CPI of 0.93, both of which fall outside of the general threshold range noted above and would accordingly be flagged for management attention under the reporting protocol that has been adopted for the Program, the composite SPI of 0.87 in particular being indicative of the Program as a whole running meaningfully behind its baseline schedule, a finding that is broadly consistent with anecdotal feedback received from several of the individual project managers regarding permitting delays and, in certain cases, extended procurement lead times for long-lead equipment items that in some instances have exceeded the lead time assumptions that were built into the original baseline schedules at the time those schedules were developed and approved, in some cases as much as eighteen to twenty-four months prior to the current reporting date, meaning that certain of the lead time assumptions that were reasonable at the time may simply no longer reflect current market conditions.

Commented [RS2]: This paragraph had some issues, especially in the back end, where it became unclear what was being referred to with the "18 to 24 months" comment. To clean it up and make it more readable, the entire paragraph has been rewritten to focus on the topic (being behind schedule and over budget) and reduce complexity. This should be much more readable. Please double check that I captured your intention correctly.

Project	Baseline Budget	PV	EV	AC	SPI	CPI	Status
Project A	\$4,200,000	\$2,100,000	\$2,050,000	\$2,010,000	0.98	1.02	On Track
Project B	\$8,750,000	\$3,900,000	\$3,760,000	\$3,850,000	0.96	0.98	Watch
Project C	\$2,100,000	\$1,050,000	\$1,040,000	\$1,015,000	0.99	1.02	On Track
Project D	\$12,400,000	\$6,800,000	\$5,120,000	\$6,240,000	0.75	0.82	At Risk
Project E	\$3,300,000	\$1,600,000	\$1,585,000	\$1,590,000	0.99	1.00	On Track
Project F	\$5,600,000	\$2,450,000	\$2,300,000	\$2,340,000	0.94	0.98	Watch
Project G	\$9,100,000	\$4,700,000	\$3,760,000	\$4,010,000	0.80	0.94	At Risk
Project H	\$1,850,000	\$1,850,000	\$1,845,000	\$1,820,000	1.00	1.01	On Track
Project I	\$6,700,000	\$2,900,000	\$2,780,000	\$2,830,000	0.96	0.98	Watch

Of the nine active projects currently being tracked under the quantitative EVM reporting protocol, ~~four two~~ are currently reflecting SPI values below the 0.90 threshold. ~~Project D and Project G, and it is worth noting that the two projects reflecting the lowest SPI values in the current reporting period, Project D and Project G, are also, not coincidentally in the assessment team's view, the~~ The two projects ~~within the current Program scope~~ that involve the greatest degree of interfaceing with the existing central utility infrastructure discussed elsewhere in facilities condition reporting. ~~suggesting This suggests~~ that schedule risk associated with utility interface and shutdown coordination may be a systemic issue affecting multiple projects ~~within the Program~~ rather than an issue isolated to any single project, a hypothesis ~~that~~ program controls staff intend to explore further in coordination with the facilities operations group.

4. Cost Performance Discussion

While the composite CPI of 0.93 is likewise outside of the target threshold range, ~~it should be noted that~~ this figure is influenced disproportionately by cost overruns on a single project, Project D, ~~(previously referenced discussed above) in the schedule performance discussion, and that if~~ Project D is excluded from the composite calculation, the resulting CPI for the remaining eight projects improves to 0.98, ~~which would fall within the acceptable threshold range. This suggests~~ing that the cost-performance concern at the portfolio level is currently concentrated rather than systemic. ~~in nature, although p~~ Program controls staff will continue to monitor the remaining projects closely, ~~given that since several of them, as discussed above, are also~~ exhibiting

unfavorable schedule performance, ~~and~~ There is a reasonably well-established relationship in program controls practice, between schedule slippage and downstream cost growth, particularly ~~with respect to extended extended general general~~ conditions costs and potential acceleration costs that may be incurred ~~in an effort~~ to recover lost ~~scheduled~~time.

5. Risk and Recommendations

Based on the findings summarized above, the following actions are recommended for consideration by Program leadership:

1. ~~e~~Convene a focused review with the management team for Project D ~~project team~~ to develop a formal recovery plan addressing both the schedule and cost variances identified on that project.
2. ~~d~~Direct program controls staff to complete the systemic risk assessment referenced above regarding utility interface coordination across Project D, Project G, and any other projects with similar interface requirements; and
3. ~~e~~Consider whether the long-lead procurement assumptions embedded in baseline schedules for projects that have not yet reached the procurement phase should be reviewed and, where warranted, updated to reflect current market lead times before those schedules are finalized, ~~in order~~ to avoid replicating the schedule risk ~~currently being experienced on Project D and Project G outlined above~~ on future Program projects ~~within the Program~~.

6. Next Steps

Program controls staff will continue to update the EVM metrics presented in this report ~~on a~~ monthly ~~basis going forward~~, and will issue a focused variance report on Project D within two weeks of ~~the date of this~~the receipt of this report, consistent with the variance reporting threshold protocol referenced in Section 2 above.

Commented [RS3]: While minimal/judicious use of bulleted and numbered lists is generally recommended, this is a perfect use case to improve clarity of the recommendations.